AI Systems Performance Engineering

Chapter 6 — GPU Architecture, CUDA Programming, and Maximizing Occupancy

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This Talk in Three Questions

1. How does a GPU turn massive parallelism into throughput?

Threads are organized into **warps and blocks** across SMs. Parallelism **executes more work at once** — and **keeps other work ready** when a warp stalls (that second part is *latency hiding*). (Part I)

2. How does CUDA expose enough parallel work?

Grids, blocks, and launch parameters decide how much work the GPU receives. (Part II)

3. Once the GPU is busy, what becomes the next bottleneck?

Usually **data movement**: the memory hierarchy, the limits of occupancy, and the roofline tell you what to fix next. (Parts III–V)

Plain English: first half: how to ***make the GPU busy****. Second half: why a busy GPU can* ***still be slow****.*

Roadmap: Five Parts, Each Answers One Question

Part	Slides	The question it answers
I GPU architecture	4–19	How is parallel work **organized and scheduled**?
II CUDA programming	20–27	How do we **map a computation onto GPU threads**?
III Memory hierarchy	28–36	Where does data **live**, and what does **movement cost**?
IV Occupancy in practice	37–45	**When** do more warps actually help?
V Correctness & roofline	46–49	Optimize **compute** next, or **memory**?

The whole chapter in one sentence

Expose enough parallelism, keep data close, and profile the real bottleneck.

When a detail page appears (SFU, TMEM, Unified Memory. . .), it always belongs to one of these five questions.

- CPUs: few cores, deep caches, fast *single* threads. GPUs: hundreds of **SMs** (streaming multiprocessors) running **thousands of threads** in parallel.
- Basic CUDA flow (right): host copies data **CPU** $\rightarrow$ **GPU**, launches a **kernel**, copies results back.
- The GPU's edge is **massive parallelism**: many SMs process different data **at once**; within an SM, **resident warps** also fill memory stalls.
- Sweet spot: **data-parallel** work — matrix multiplies, convolutions — written in CUDA C++ or via **PyTorch** / OpenAI **Triton**.[1,3]

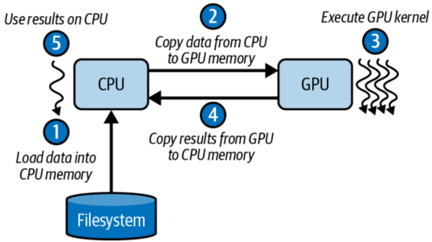


Figure 6-1. Simple CUDA programming flow

Plain English: the CPU finishes one part fast; the GPU is a **factory** *— keep* **every workshop busy***.*

The Thread Hierarchy: Threads → Blocks → Grids

- **Thread**: runs your kernel code on one data element.
- **Thread block** (CTA — cooperative thread array): up to **1,024 threads**; shares fast on-chip **shared memory**.
- **Grid**: all blocks of one launch — sized right, it scales to **millions of threads** with zero kernel changes.
- The CUDA runtime (and PyTorch) handles scheduling and distribution across all SMs.

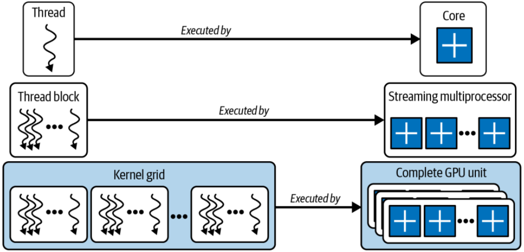


Figure 6-3. Threads, thread blocks (CTAs), and grids

Plain English: workers (threads) form teams (blocks) that talk cheaply inside the team; the company (grid) hires as many teams as the job needs.

Warps and SIMT: 32 Threads in Lockstep

- Blocks are subdivided into **warps of 32 threads** that execute one instruction together under **SIMT** (single instruction, multiple threads).
- The warp — not the thread — is what the **hardware actually schedules**; warp size has been **32 on every GPU generation**.
- The hardware hides long-latency events (global loads, cache fills, pipeline stalls) by **rapidly switching among warps**.

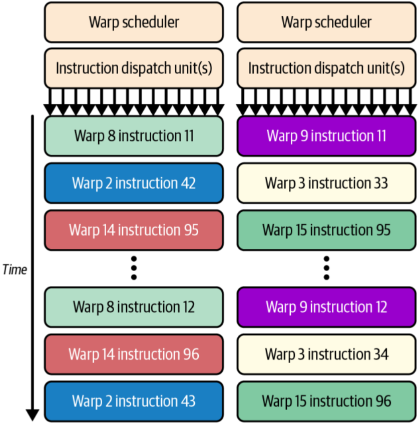


Figure 6-7. A warp advances as a whole under the warp scheduler

Plain English: a warp is a **32-worker crew***: one instruction for all 32, everyone moves together — or the whole crew waits.*

Inside a Blackwell SM: The Resource Budget

- Each SM tracks up to **64 warps** = **2,048 threads** concurrently — that's the pool the scheduler switches between.
- **64K 32-bit registers** per SM (256 KB total); any single thread can use at most **255**.
- **256 KB** combined L1 cache / shared memory; up to **228 KB** configurable as user-managed shared memory (**227 KB usable** — CUDA reserves 1 KB per block).
- These on-chip budgets let one SM juggle thousands of threads without constantly touching DRAM.[2]

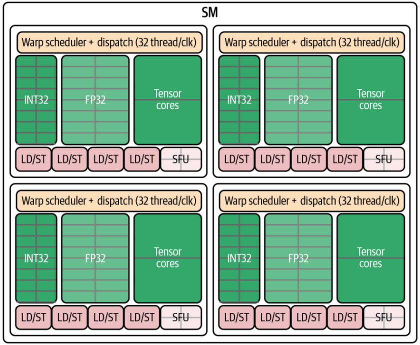
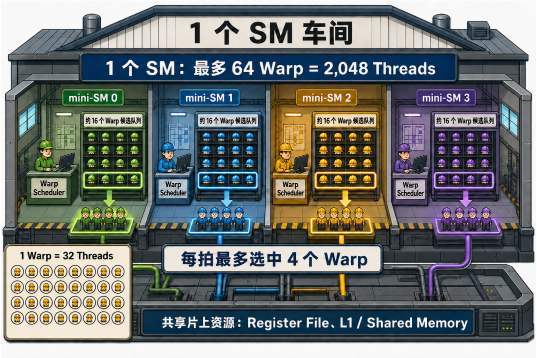


Figure 6-2. A Blackwell SM: four scheduling partitions sharing on-chip resources

Plain English: an SM is a workshop: registers are toolbelts, shared memory is the workbench.

Analogy: One SM Is a Workshop



Presenter's analogy figure (not from the book)

- Four bays = **4 scheduling partitions** ("mini-SMs"), one **warp scheduler** each — CUDA only ever sees the whole SM.
- Hard hats on shelves = **resident warps**: ∼16 per partition, **64 total**. Resident = state already allocated — *not* "executing right now."
- Highlighted crews = the **≤4 warps issued this cycle** (each scheduler picks at most 1 *ready* warp).
- Bottom left: **1 warp = 32 threads** — dispatched as a whole crew.
- Common foundation: **register file, L1, shared memory** — shared by all four partitions.

Plain English: 64 crews on standby, 4 dispatchers, at most 4 crews per beat — each crew is 32 workers.

Let the Workshop Move: Latency Hiding, Cycle by Cycle

One partition, one moment

Warp A — **waiting** on an HBM load
Warp B — **waiting** on a previous result
Warp C — **ready**   Warp D — **ready**

The scheduler skips A and B and **issues C or D** — skipping a stalled warp costs **nothing**.

Cycle	Issued (example)
1	Warp C
2	Warp F
3	Warp D
4	Warp H

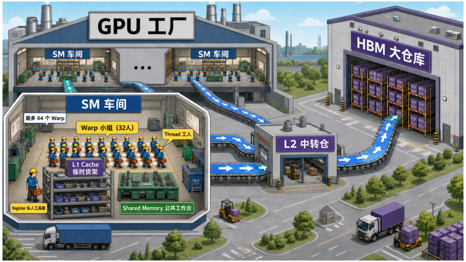
- "Issued" = the instruction **starts**; it need not finish this cycle.
- Latency hiding does **not** shorten any warp's wait — it **fills the wait** with other warps' work.
- This is why **64 resident warps** matter: a deep pool of *ready* candidates for the 4 dispatchers — and why low occupancy hurts (nothing ready to switch to).

Plain English: latency hiding does ***not*** *reduce waiting time — it* ***fills the waiting time*** *with other crews' work.*

Two Views of One Machine: Software vs. Hardware

Software (you write)	**Hardware (runs it)**	**Analogy**
Grid	the whole GPU	the factory
Block	exactly **one SM**	one workshop
(warp: implicit)	unit of scheduling	32-person crew
Thread	one lane of a warp	one worker

- You choose **grid and block sizes**; the hardware decides **block placement and warp scheduling**.
- The **warp is the bridge**: hardware slices every block into warps of 32 — automatically, invisibly.
- Memory preview (Part III): registers = toolbox, shared = workbench, L1 = shop shelf, **L2 = central depot**, **HBM = warehouse**; a global load tries L1 $\rightarrow$ L2 $\rightarrow$ HBM and returns the same way.



Presenter's analogy: SM workshops, L2 depot, HBM warehouse

Plain English: software speaks in grids and blocks; hardware answers in SMs and warps — the warp is where the two vocabularies meet.

Four Warp Schedulers, Dual Issue

- Each SM = four "**mini-SMs**": independent warp schedulers with their own dispatch logic.
- Each scheduler issues **1 warp instruction per cycle** ⇒ up to **4 warps advance per cycle**.
- **Dual-issue**: one math (INT32/FP32/Tensor Core) + one memory (load/store) instruction per cycle — *from the same warp*, never across warps.
- Best case: **4 math + 4 memory** instructions in flight every cycle.

Per cycle (Table 6-1)	Limit
Schedulers	4
Warps issued	4
Math / memory ops	4 / 4

Book note: table values are illustrative; real benchmarks in the GitHub repo[7]

Plain English: **four dispatchers** *on one floor, each sending out one crew per beat — and a crew can compute with one hand while fetching parts with the other.*

SFUs and Load/Store Pipelines

SFU — Special Function Unit

- Handles **transcendentals**: sine, cosine, reciprocal, square root.
- Own **dedicated pipeline** — *not* part of the dual-issue math+memory pair.
- Slow functions never stall the core compute/memory pipes ⇒ more **instruction-level parallelism** for mixed-operation kernels.

LD/ST — Load/Store pipelines

- **16 LD/ST pipelines** per SM (4 per scheduler) feed L1/shared, L2, and global memory.
- **Caveat (book warning):** exact counts and pairings are **not guaranteed** — rely on profiling counters and the **Blackwell tuning guide**.[2]

Plain English: the SFU is a specialist counter at the back of the store — complicated requests go there so the express lanes keep moving.

Blocks Cooperate Inside, Stay Independent Outside

- Within a block: share data via shared memory, synchronize with `__syncthreads()` (a barrier: everyone arrives before anyone continues).
- **Every barrier costs time** — the book's advice: **minimize synchronization points**.
- Across blocks: **independent, no guaranteed order**. That freedom lets the scheduler spray blocks over all SMs — and lets your kernel run **unmodified on future GPUs** with more SMs.

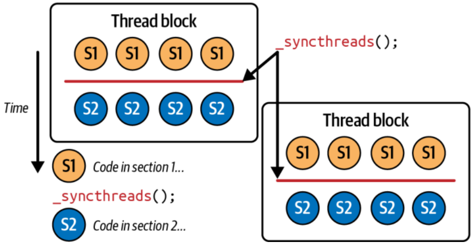


Figure 6-6. `__syncthreads()` between two code sections

Plain English: team huddles (barriers) are useful but expensive — call as few as possible; and never assume team A finishes before team B.

Thread Block Clusters and DSMEM

- Traditionally, threads in *different* blocks could not cooperate. Modern GPUs add **thread block clusters**: blocks that communicate **across SMs**, with cluster-scoped hardware barriers.
- Backed by **DSMEM** (distributed shared memory): the shared-memory banks of participating SMs, linked over a fast **on-chip interconnect** into one pool.
- Threads read, write, and atomically update each other's shared buffers **at on-chip speeds — without global-memory bandwidth**.
- Key enabler for huge matmuls in LLMs — details in **Chapter 10**.

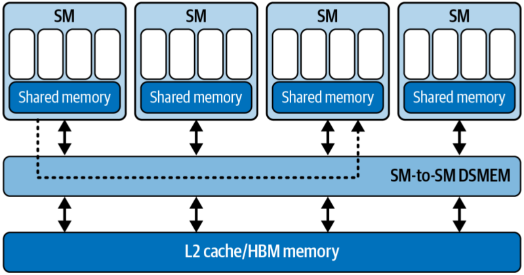


Figure 6-5. DSMEM shared across thread blocks in a cluster

Plain English: neighboring teams knock a door through the wall between their workbenches instead of mailing parts through the warehouse.

Occupancy Gives the Scheduler More Choices

Definition

Occupancy = active warps on an SM ÷ the hardware maximum (64 on Blackwell).
"**Achieved occupancy**" is the measured average, reported by Nsight Compute.

- High occupancy ⇒ when one warp stalls on memory, another is ready ⇒ **latency hiding**.
- Blackwell's large register file (64K/SM) makes high occupancy easier to reach.
- The counterweight: warps share **registers and shared memory**. Pack in too many ⇒ **register spilling** to slow memory — a new stall you created.
- So: profile occupancy **together with** register / shared-memory usage.

Plain English: occupancy is **not** *"how busy the GPU is" and* **not** *a performance score — it counts* **how many resident crews the dispatcher can choose from***. Enough hides latency; beyond enough, more may not help* *(p. 51)**.*

Warp Divergence: When `if/else` Splits the Crew

- If threads *within one warp* take different branches, the warp **serializes**: it runs the `if` path with the other lanes **masked**, then the `else` path.
- Execution time multiplies by the **number of distinct paths**.
- **No penalty across warps** — different warps may branch differently for free.
- Practical corollary: branch on **warp-aligned** conditions (`threadIdx.x / 32`), not per-thread parity (`threadIdx.x % 2`).
- Detection and mitigation: Chapter 8 (Nsight Compute).

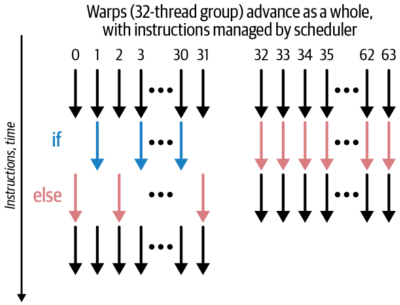


Figure 6-8. SIMT warp divergence (left) vs. uniformity (right)

Plain English: if half the **crew** *takes one work order and half another, the crew runs* **each job twice**.

Hardware Limits I: Warp and Block (Table 6-2)

Don't memorize these numbers (this page and the next). The one idea: each block's threads, registers, and shared memory decide how many blocks and warps still fit on one SM.

Resource	**Limit (B200)**
Warp size	32 threads
Max threads / block	1,024
Max warps / block	32 $(= 1024 \div 32)$

`blockDim.x` $\times$ `blockDim.y` $\times$ `blockDim.z` $\leq$ `1024`

Why multiples of 32 matter

A **33-thread block** occupies **two** warp slots — the second warp runs **1 of 32 lanes** but still consumes a full scheduler slot. Every non-multiple of 32 donates scheduler capacity to the void.

- Block too big $\Rightarrow$ too many registers per block $\Rightarrow$ **spilling**; or too much shared memory (227 KB usable is shared by **all resident blocks** on the SM).
- Smaller blocks can mean **higher occupancy** — more independent blocks fit per SM.

Hardware Limits II: SM-Resident and Grid (Tables 6-3 / 6-4)

Per SM (resident)	**Limit**
Max warps	**64** (= 2,048 threads)
Max threads	2,048
Max blocks	32
Grid	
Max blocks (X / Y / Z)	∼2.1B / 65,535 / 65,535
Max concurrent grids	**128** kernels per device

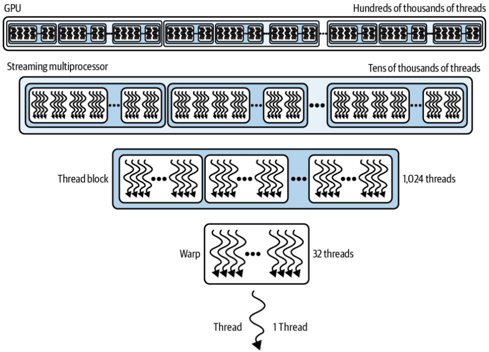


Figure 6-9. Relative scale of thread resources

- Worked example: 1024-thread blocks ⇒ only **2 blocks/SM**; 256-thread blocks ⇒ **8 blocks/SM** — same 2,048 threads, more flexibility.
- In practice you hit **per-SM limits** long before grid limits; >65,535 blocks in Y/Z ⇒ use 2D/3D grids or multilaunch.[1,2]

Compatibility: PTX, SASS, and Fatbins

- **SASS**: final machine code for *one* architecture (sm_90 = Hopper, sm_100 = Blackwell). SASS-only binaries **won't run on newer GPUs**.
- **PTX**: virtual ISA the driver **JIT-compiles** at load time ⇒ **forward compatibility**.
- Family targets (`sm_100f`) restrict portability to the same feature family.

Best practice

Ship a **fatbin**: generic PTX + the family-specific cubins you need for optimizations — with **fallback paths** for other architectures.

- Verify with `CUDA_FORCE_PTX_JIT=1`: forces PTX JIT; if the binary lacks PTX, **kernel launch fails** — rebuild with PTX.

Plain English: engineering takeaway: fast code must also **survive GPU generations** *— ship optimized* **SASS** *for today's chips plus* **PTX** *for forward compatibility. (40 seconds, then back to the main line.)*

Part II — Anatomy of a CUDA Kernel: Device Side

Bridge: the hardware wants many ready warps — now: how does CUDA code create them?

```
__global__ void myKernel(float* input, int N) {
    // unique global index for this thread
    int idx = blockIdx.x * blockDim.x + threadIdx.x;
    if (idx < N) { // bounds check!
        input[idx] *= 2.0f; // in-place: double it
    }
}
```

- `__global__`: runs on the device, callable from the host.
- Built-ins: `blockIdx` (which block am I), `blockDim` (block size), `threadIdx` (my rank in the block).

- The kernel is written from the perspective of **one thread**; `idx` carves out its slice of the data.
- CUDA compiles it into device code executed by **thousands or millions** of lightweight threads in parallel.
- Launch syntax (host): `<<<blocksPerGrid, threadsPerBlock>>>` — the two core parameters of *every* kernel invocation.

Plain English: write the job description for one worker; CUDA prints a million copies, each with a different row number.

Host Side: The Six-Step Data Flow

```
const int N = 1'000'000;
float *h_input = nullptr, *d_input = nullptr; // h_=host, d_=device
cudaMallocHost(&h_input, N * sizeof(float)); // 1) pinned host alloc
for (int i = 0; i < N; ++i) h_input[i] = 1.0f; // init
cudaMalloc(&d_input, N * sizeof(float)); // device alloc
cudaMemcpy(d_input, h_input, N * sizeof(float),
          cudaMemcpyHostToDevice); // 2) H2D copy
const int threadsPerBlock = 256;
const int blocksPerGrid = // = 3,907 for 1M
    (N + threadsPerBlock - 1) / threadsPerBlock;
myKernel<<<blocksPerGrid, threadsPerBlock>>>(d_input, N); // 3) launch
cudaDeviceSynchronize(); // 4) wait for device
cudaMemcpy(h_input, d_input, N * sizeof(float),
          cudaMemcpyDeviceToHost); // 5) D2H copy
cudaFree(d_input); cudaFreeHost(h_input); // 6) cleanup
```

- Naming convention: `h_` = host pointer, `d_` = device pointer — used throughout the book.
- `cudaMallocHost` = **pinned** (page-locked) memory — prerequisite for async copies to truly overlap later.
- Book note: *not fully optimized yet* — this is the simple, complete **template** the rest of the book improves on.

Why Pass N? The Single-Thread Perspective

The book's answer

"A CUDA kernel function is designed to work **inside of a single thread**, alongside thousands of other threads, **on a partition of the input data**. As such, N defines the size of the partition that this particular kernel will process."

- A CPU function could just inspect the array size. A kernel **cannot** — it sees a raw pointer and must be told where its world ends.
- Combined with `blockIdx`/`blockDim`/`threadIdx`, N lets every element be processed **cleanly and uniquely**, in parallel, across many SMs.
- This is the core mental shift from CPU to GPU programming: you don't write "process the array," you write "process **my** element — if it exists."

Plain English: each of the million copied workers gets the same memo; N is the line that says where the job ends.

The Bounds Check — and Lazily Surfacing Errors

Worked example: $N = 63$

Scheduler assigns **2 warps** (64 threads). Warp 1 handles elements 0–31 — fine. Warp 2 expects 32–63 **plus one extra thread**; without `if (idx < N)` it reads past the array ⇒ `cudaErrorIllegalAddress`.

- Book rule: bounds checks are everywhere in CUDA; "if you don't see it, **understand why it's not there**."

- Kernels run **asynchronously, with no per-thread exceptions**: an illegal access sets a **global fault flag** for the whole launch.
- The host only sees it at the **next sync or CUDA API call** — errors surface **lazily**.
- Practice: `cudaGetLastError()` + `cudaDeviceSynchronize()` right after launches, so faults are caught where they happen.

Plain English: the GPU doesn't call you when something breaks — it leaves a note you only find the next time you check the mailbox.

Choosing Launch Parameters: The 256 Recipe

Recipe that works almost everywhere

Start with `threadsPerBlock = 256` (8 warps); `blocksPerGrid = (N + 255) / 256` (round up — every element covered). Tune 128/512 by profiling.

- **Multiple of 32**: no half-empty warps occupying scheduler slots.
- **Latency hiding**: hundreds of threads per SM cover DRAM and instruction stalls — 8 blocks $\times$ 256 $=$ 2,048 *can* fill an SM, **if registers and shared memory allow**.
- **Occupancy**: 8 warps/block usually fits registers and shared memory comfortably.
- **Resource-balanced**: far from the 1,024 cap, big enough to keep warps busy.
- Book tip for Blackwell: consider **256–512** threads per block.

Plain English: 256 is the "medium coffee" of block sizes — almost never the wrong first order, refine later if profiling says so.

2D and 3D Kernel Inputs

```
__global__ void my2DKernel(float* input,
                           int width, int height) {
    int x = blockIdx.x * blockDim.x + threadIdx.x;
    int y = blockIdx.y * blockDim.y + threadIdx.y;
    if (x < width && y < height) { // 2D bounds check
        int idx = y * width + x; // flatten to 1D
        input[idx] *= 2.0f;
    }
}
// host: dim3 threads(16,16); // 256 threads, 2D
// dim3 blocks((W+15)/16, (H+15)/16);
```

- Natural fit for **images**: e.g., a 1,024 × 1,024 matrix processed by 16 × 16 blocks (still 256 threads).
- Same pattern generalizes to 3D with `dim3(x, y, z)` — volumetric data maps directly onto the thread hierarchy.
- Book note: most of the book uses **1D or 2D (tiled)** launches; in 1D, plain ints beat `dim3`.

Plain English: same recipe, two axes: each worker gets a (row, column) badge instead of a single row number.

Allocate Asynchronously: Streams and Memory Pools

```
cudaStream_t s;
cudaStreamCreateWithFlags(&s, cudaStreamNonBlocking);

float* d_buf = nullptr;
cudaMallocAsync(&d_buf, N * sizeof(float), s);
myKernel<<<blocks, threads, 0, s>>>(d_buf, N);
cudaFreeAsync(d_buf, s); // deferred until s finishes
```

- **Free waits only for stream `s`** — no global `cudaDeviceSynchronize`, no cross-stream sync.

- `cudaMalloc`/`cudaFree` are **synchronous and expensive**: full-device sync + OS calls (`mmap`/`ioctl`), kernel-space context switches.
- `cudaMallocAsync`/`cudaFreeAsync` draw from a per-device **memory pool**: recycled buffers, no OS round trips, **less fragmentation** over long training loops.
- Non-blocking streams avoid legacy **default-stream barriers** (book tip; multistream overlap in Chapter 11).

Plain English: keep a truck fleet in the lot and grab keys — don't buy and return a truck for every delivery.

Tuning the Pool; PyTorch's Caching Allocator

Pool knobs

- `cudaMemPoolAttrReleaseThreshold` (via `cudaMemPoolSetAttribute`): how much reserved memory the pool keeps before releasing to the OS.
- `cudaMemPoolTrimTo`: proactively return memory.
- Trade-off: total **footprint** vs. **fragmentation**.

The PyTorch connection

- PyTorch's **caching allocator** (`PYTORCH_ALLOC_CONF`, formerly `PYTORCH_CUDA_ALLOC_CONF`) is the same idea: reuse GPU memory, avoid a synchronous `cudaMalloc` per new tensor per iteration.

- Verdict from the book: one-off buffers — blocking `cudaMalloc` is fine; **allocation-heavy loops — async + pools** give more consistent performance and higher throughput.

Part III — The Memory Ladder at a Glance

Bridge: we can now create plenty of threads — but what are they all waiting for? Usually data.

Level	Scope	Capacity	Latency	BW
Registers	thread	256 KB / SM	~1 cyc	10s TB/s
Shared + L1	SM	228 KB usable	20–30 cyc	TB/s
TMEM	SM	256 KB (TC)	~10 cyc	TB/s
Const cache	SM	8 of 64 KB	1 cyc bcast	TB/s
L2 cache	GPU	**126 MB**	~200 cyc	multi-TB/s
Local (spill)	thread	DRAM-backed	≤1000 cyc	—
Global HBM3e	device	**180 GB** (B200)	≤1000 cyc	**~8 TB/s**

Table 6-5. Every level trades capacity for latency/bandwidth.

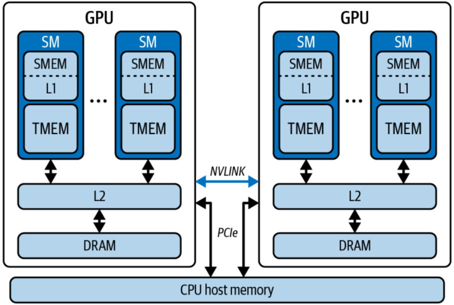


Figure 6-10. GPU memory hierarchy, including the CPU

Plain English: your **own toolbox** *(registers) → the* **shared workbench** *(shared) → the factory's* **central depot** *(L2) → the* **distant warehouse** *(HBM) — do the work at your bench.*

Register Pressure Can Turn Fast Storage into DRAM Access

- Every thread "begins its journey" at the **register file**: single-cycle reads/writes, contending with almost nothing — **tens of TB/s per SM**.
- Budget: 64K registers per SM, **max 255 per thread**.
- The cliff: need more (too many locals, compiler temporaries) $\Rightarrow$ overflow **spills into local memory** — which despite the name lives in **off-chip DRAM**: hundreds to 1000+ cycles.
- Watch **"Registers Per Thread"** in Nsight Compute — spills are silent killers.

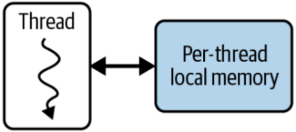


Figure 6-12. Per-thread local memory (spill space, DRAM-backed)

Plain English: "local memory" is a self-storage unit across town, not your pocket.

Shared Memory + L1: One SRAM, Two Jobs

- One **256 KB SRAM per SM**, split between user-managed **shared memory** (up to 228 KB; 227 usable per block) and **L1/data cache**.
- You choose the **carveout**:

```
cudaFuncSetAttribute(myKernel,
  cudaFuncAttributePreferredSharedMemoryCarveout,
  carveoutPercent);
```

- ∼20–30 cycles; **TB/s** if you avoid **bank conflicts** (multiple threads hitting the same memory bank ⇒ serialized access).
- This is the workbench for **intrablock cooperation** — tiles, reductions, staging.

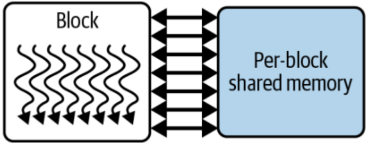


Figure 6-13. Thread-block shared memory

Plain English: one workbench, adjustable split: how much is your team's project table vs. the automatic cache shelf.

TMEM and TMA: Feeding the Tensor Cores

- **TMEM**: dedicated **256 KB per-SM SRAM**, the **accumulator** for 5th-gen Tensor Core ops (`tcgen05.*`, **UMMA**); talks to Tensor Cores at tens of TB/s.
- **Not pointer-addressable** from CUDA C++ — data movement is orchestrated by the **TMA** (Tensor Memory Accelerator) via **descriptors**.
- For $C = A \times B$: operand B in SMEM, A + accumulator in TMEM; TMA moves tiles HBM $\rightarrow$ L2 $\rightarrow$ SMEM; SMEM $\leftrightarrow$ TMEM implicit via Tensor Core ops.
- Net effect: **reduces Tensor Core reliance on global memory**. Details: Chapter 10.

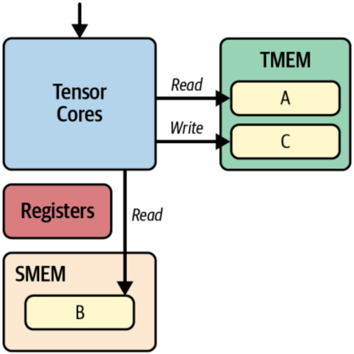


Figure 6-11. TMEM + SMEM servicing Tensor Cores for $C = A \times B$

Plain English: TMA is a forklift crew moving pallets in the background; the chefs never leave the kitchen.

Constant Cache: One-Cycle Broadcast

- ∼**8 KB cache per SM** fronting the 64 KB `__constant__` space.
- Superpower: when all 32 threads of a warp load **the same address**, the cache **broadcasts** the value in a **single cycle** — as fast as a register.
- Divergent reads **serialize across lanes**; misses cost more — so it's for **small, read-only, uniformly-accessed** data.

The book's LLM examples

- Rotary positional encoding (RoPE) lookup tables
- ALiBi slopes
- LayerNorm γ/β vectors
- Embedding quantization scales

Shared by every thread with **zero global-memory traffic**.

Plain English: a megaphone, not a mailbox: one announcement reaches all 32 workers at once — but only if they all asked the same question.

L2 Cache: The GPU-Wide Middleman

- **126 MB** shared by all SMs — the glue between on-chip SRAM and off-chip HBM3e.
- $\sim$200 cycles, aggregate bandwidth in the **tens of TB/s**; absorbs L1 spillover.
- **Cross-block reuse**: data fetched by one thread block is reused by others **without revisiting DRAM**.

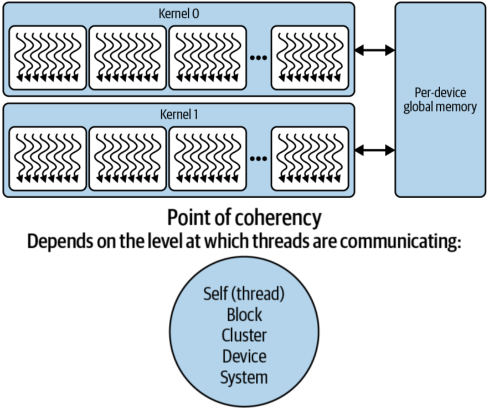
The coalescing rule (book tip)

Structure global loads into **128-byte aligned, coalesced** segments that map cleanly to cache lines — avoids split transactions, maximizes L2 **and** DRAM bandwidth. (How-to: Chapter 7.)

Plain English: L2 is the factory's **central depot***: if another crew already fetched the part from the warehouse, you grab it here — provided everyone orders* **whole pallets***, not single screws.*

Global HBM3e, the Dual-Die B200, and Coherency

- **180 GB** on B200 (∼288 GB on B300) at **∼8 TB/s** — but **hundreds to 1000+ cycles**: the slowest link in the chain.
- Register spills and oversized automatic arrays (local memory) also pay this price.
- **Fun fact (book note)**: B200 = **two reticle-limited dies** joined by a **10 TB/s** chip-to-chip interconnect, 4 HBM3e stacks each — presented to you as **one GPU, one address space**.
- **Point of coherency** (Figure 6-15): memory consistency is established at thread → block → cluster → device → system scope — the wider the communication, the higher the cost.



Figures 6-14 / 6-15. Global memory; points of coherency

Unified Memory Hides Copies, Not Their Cost

- `cudaMallocManaged()`: one coherent CPU+GPU address space — no separate buffers, no manual `cudaMemcpy`.
- Under the hood: pages **migrate on demand** over whatever link connects CPU and GPU.
- The catch: a GPU touch of a CPU-resident page **page-faults and stalls the kernel**.
- On PCIe: on-fault transfers can be **slower than manual memcpy**. On Grace superchips, **NVLink-C2C** (∼900 GB/s) makes migrations near device-native — **but latency is never zero**.

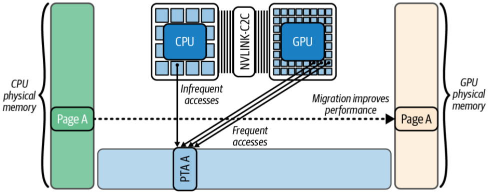


Figure 6-16. Automatic page migration with Unified Memory

Plain English: Unified Memory is **automatic parts delivery***: convenient — but if the part isn't in the workshop when the crew needs it,* **the crew stands and waits***.*

Taming Unified Memory: Prefetch, Advise, Attach

```
// 1) move data BEFORE the kernel needs it:
cudaMemPrefetchAsync(ptr, size, gpuId, s);
// 2) tell the driver how you'll use it:
cudaMemAdvise(ptr, size,
    cudaMemAdviseSetPreferredLocation, gpuId);
cudaMemAdvise(ptr, size,
    cudaMemAdviseSetReadMostly, gpuId);
cudaMemAdvise(ptr, size,
    cudaMemAdviseSetAccessedBy, otherGpuId);
// 3) confine faults/migrations to one stream:
cudaStreamAttachMemAsync(stream, ptr, 0,
    cudaMemAttachSingle);
```

- **Prefetch** turns first-touch migrations into **overlappable async transfers** (Figure 6-17).
- **ReadMostly** allows replicated read-only copies; **AccessedBy** lets a second GPU map pages **without** migration (Figure 6-18).
- **Attach** stops *other* streams from stalling on this range.
- No NVLink-C2C? Pin data **NUMA-local** (peer copy / prefetch).
- Result: performance **close to manual** `cudaMemcpy`, simplicity retained.

Part IV — Occupancy Ground Rules

Bridge: the memory ladder explains why warps stall. Occupancy asks: how many other warps do we need to fill those gaps?

The most fundamental rule of CUDA performance (book)

"Launch enough parallel work to fully occupy the GPU."

Rule 1 — low occupancy, poor perf

First remedy: **increase parallelism** (more blocks/threads) until there are **enough ready warps to hide latency** — stop when performance **plateaus**.

Rule 2 — occupancy already high

If the kernel is **memory-bound**, pushing to 100% won't help: you need **just enough warps to hide latency** — beyond that, the bottleneck is bandwidth.

- Up next: one operation ($C = A + B$, $N = 10^6$), two implementations, and what the profilers say about them.

Plain English: rule 1 **fills the workshop with crews***; rule 2: a full workshop* **still stalls** *if the warehouse truck is the bottleneck.*

Case Study, Take 1: addSequential

```
// Single thread does ALL N additions
__global__ void addSequential(const float* A,
    const float* B, float* C, int N) {
  if (blockIdx.x == 0 && threadIdx.x == 0) {
    for (int i = 0; i < N; ++i) {
      C[i] = A[i] + B[i];
    }
  }
}
// Note: this kernel assumes <<<1,1>>>.
addSequential<<<1,1>>>(d_A, d_B, d_C, N);
cudaDeviceSynchronize();
```

- One thread, one warp, one SM — **everything else watches**.
- Book: "the GPU's vast resources are mostly idle... very poor occupancy and, ultimately, low performance."
- No other warps to switch to $\Rightarrow$ memory waits become **pure idle time** — zero latency hiding.

Plain English: you rented the whole factory and hired one worker.

The Same Trap in PyTorch

```
import torch
N = 1_000_000
A = torch.arange(N, dtype=torch.float32,
                 device='cuda')
B = 2 * A
C = torch.empty_like(A)
torch.cuda.synchronize()
# Naive, sequential GPU ops - DO NOT DO THIS
with torch.inference_mode():
    for i in range(N): # N tiny serial kernels!
        C[i] = A[i] + B[i]
torch.cuda.synchronize()
```

- A Python loop around GPU ops **serializes N tiny kernel launches** — the GPU becomes “a scalar, nonparallel processor.”
- Same terrible occupancy, plus **per-launch CPU overhead** $\times$ 1,000,000.
- Book advice: there is **almost always an optimized PyTorch-native (vectorized) implementation** — including compiler-emitted code.

Plain English: the most common GPU bug: accidentally using the GPU as a very expensive single-core CPU.

Case Study, Take 2: addParallel — and `C = A + B`

```
// One thread per element
__global__ void addParallel(
    const float* __restrict__ A,
    const float* __restrict__ B,
    float* __restrict__ C, int N) {
  int idx = blockIdx.x*blockDim.x + threadIdx.x;
  if (idx < N) C[idx] = A[idx] + B[idx];
}
addParallel<<<(N+255)/256, 256>>>(dA, dB, dC, N);
```

The book's full version also uses pinned host memory, a non-blocking stream, and `cudaMallocAsync`/`cudaMemcpyAsync` — the habits from Part II.

- $\sim$3,907 blocks $\times$ 256 threads — **a million threads** cover the array (Figure 6-19).
- `__restrict__`: promises no pointer aliasing — frees the compiler to optimize.
- PyTorch equivalent is one line: `C = A + B` — a single vectorized kernel engaging many threads concurrently.

Plain English: same factory, now every workstation is staffed — and the PyTorch version is just "hire the staffing agency."

Measuring It: nsys and ncu

```
# Nsight Systems: whole-app timeline
nsys profile --stats=true -t cuda,nvtx \
    -o report ./add_parallel.py

# Nsight Compute: per-kernel counters
ncu --section SpeedOfLight \
   --metrics sm__warps_active.avg.\
pct_of_peak_sustained_active,gpu__time_duration.avg \
   --target-processes all \
   --print-summary per-gpu -o report ./add_parallel.py
```

- **nsys** answers "**where does time go** — is the GPU starved or blocked?"
- **ncu** answers "**why is this kernel slow** — occupancy? stalls? cache misses?"
- Run **both**: nsys alone misses in-kernel inefficiency; ncu alone can't see whether kernels are **fed data fast enough**.
- That long `sm__warps_active...` metric **is** achieved occupancy.

Plain English: nsys is the stopwatch for the whole machine; ncu is the microscope on one kernel.

Parallelism Cuts Runtime 22× at Only 38.7% Occupancy

Metric	Sequential	Parallel
Kernel time (ms)	48.21	**2.17**
GPU utilization	1.5%	**95%**
Achieved occupancy	1.3%	**38.7%**
Warp exec. efficiency	3.1%	**100%**

Table 6-6 (illustrative — real benchmarks in the book's repo[7]). Warp efficiency 3.1% ≈ 1/32: one active lane per warp.

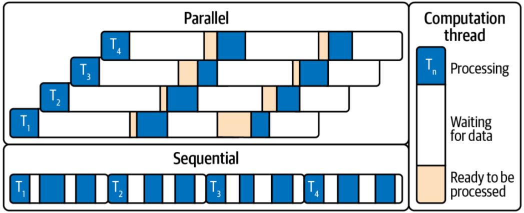


Figure 6-20. Sequential timeline has idle gaps; in the parallel version other warps' work fills them

- Note: 38.7% occupancy already bought 22× — you don't need 100%.
- GPU utilization, SM active %, achieved occupancy, and warp efficiency are **related but different metrics** — which is exactly how **95%**, **38.7%**, and **100%** coexist on one kernel.

A Busy GPU Can Still Be Waiting on Memory (LLM Decode)

Bridge: 95% utilization at only 38.7% occupancy, already 22× faster — so we don't need 100%. But are we at peak FLOPS? No: vector add mostly moves data.

- After parallelism, the next lever is per-warp efficiency (ILP etc., Chapter 8). But: **even at 100% occupancy**, performance suffers if the kernel is **memory bound** — limited by data movement, not compute.
- The book's canonical example: the **decode phase of an LLM** — every generated token streams **model weights** from HBM into registers/shared memory.
- Hundreds of billions of parameters $\times \sim 1$ byte $\approx$ **hundreds of GB** per pass — this **saturates memory bandwidth** regardless of thread count.

The trend that makes this worse (book note)

GPU FLOPS are outpacing memory bandwidth. Blackwell HBM3e delivers ∼8 TB/s, but compute and model sizes grow faster — **optimizing memory movement is absolutely critical** in modern AI workloads.

Plain English: when the whole job is "carry boxes from the warehouse," hiring more clerks at the desk doesn't help — that's what the roofline model (Part V) formalizes.

__launch_bounds__: Compile-Time Occupancy Control

```
// promise: never launched with > 256 thr/block
// request: keep >= 16 blocks resident per SM
__global__ __launch_bounds__(256, 16)
void myKernel(...) { /* ... */ }
```

- $16 \times 256 = 4{,}096 > 2{,}048$ limit $\Rightarrow$ compiler **clamps** to 8 blocks and warns:
  `ptxas warning: ... .minnctapersm will be ignored`

- Effect: compiler **caps per-thread registers** (and may restrict unrolling/inlining) to avoid spills and fit **more warps per SM**.
- The deal: trade a bit of **per-thread performance** for **more consistent warp throughput** (more warps in flight).
- Danger zone: forcing too many threads $\Rightarrow$ registers run out $\Rightarrow$ **spilling to local memory** — slower than what you started with.

Plain English: you're trading each worker's toolbox size for the number of workers on the floor — the sweet spot is empirical.

The Occupancy API: Runtime Autotuning

```
int minGridSize = 0, bestBlockSize = 0;
// if kernel uses extern __shared__, set truthfully:
size_t dynSmemBytes = 0;
cudaOccupancyMaxPotentialBlockSize(
    &minGridSize, &bestBlockSize, myKernel,
    dynSmemBytes, /*blockSizeLimit=*/0);
// cover N, but don't go below the min grid
// that saturates occupancy:
int gridSize = std::max(minGridSize,
    (N + bestBlockSize - 1) / bestBlockSize);
myKernel<<<gridSize, bestBlockSize,
          dynSmemBytes>>>(...);
```

- Computes the block size that **maximizes occupancy** given the kernel's actual register/shared-memory usage.
- Two gotchas (book): `minGridSize` saturates occupancy — it is **not** the grid that covers N (take the `max`); and **pass real dynamic shared-memory bytes**.
- Book tip: **validate with $\pm$1–2 candidate block sizes** — register pressure and L2 behavior can make slightly **sub-maximal occupancy faster**.

Part V — Debugging Correctness: NVIDIA Compute Sanitizer

Bridge: no more blind-tuning block sizes — next: compute ceiling or memory ceiling? (roofline, after a correctness detour)

Memory

- **memcheck** — out-of-bounds / misaligned accesses (global, local, shared), HW exceptions, device leaks; `--check-device-heap`.
- **initcheck** — reads of uninitialized global memory (classic cause: a **missing H2D copy**).

Concurrency

- **racecheck** — shared-memory hazards: **WAW / WAR / RAW** ⇒ nondeterministic behavior.
- **synccheck** — invalid sync primitives, mismatched barriers ⇒ deadlocks, inconsistent state.

- CLI: `compute-sanitizer [--tool <name>] <app>`; filter with `--kernel-name`, annotate with **NVTX** (`--nvtx yes`).
- Book tip: run it in **CI** with `--error-exitcode`.[5]

Plain English: "it works on my run" means nothing at 100,000 threads — the sanitizer is your seatbelt.

Roofline Tells Us Which Resource Is the Limit

- **Arithmetic intensity** = FLOPs per byte moved to/from HBM.
- Two ceilings: **flat** compute roof (~80 TFLOP/s FP32) and **sloped** memory roof (~8 TB/s), crossing at the **ridge point** ≈ **10 FLOPs/byte** (80T ÷ 8T).
- Worked example: $C = A + B$ loads 8 B, adds once, stores 4 B ⇒ 1 FLOP / 12 B = **0.083 FLOPs/byte** — >100× left of the ridge: hopelessly **memory-bound**.
- Book note: LLMs are **both** — **prefill** tends compute-bound, **decode** tends memory-bound (Ch. 15–18).

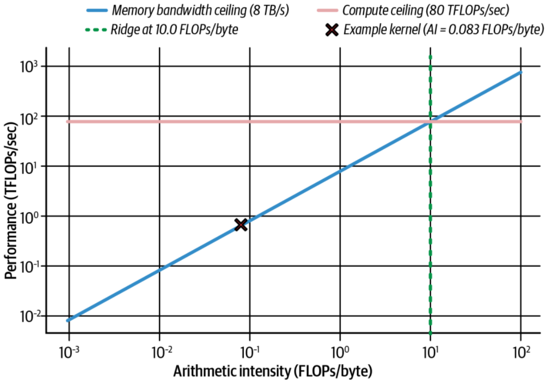


Figure 6-21. Roofline for a Blackwell-class GPU; our kernel sits at 0.083 FLOPs/byte

Plain English: the roofline asks first: is this kernel ***starving for math, or for data****?*[4]

Moving Right on the Roofline: Lower Precision Pays Twice

- To escape the memory-bound region, **do more work per byte**: reuse data on-chip, fuse ops — or simply **shrink the bytes**.
- One 128-byte transaction carries **32 FP32 = 64 FP16 = 128 FP8 = 256 FP4** values. FP32 $\rightarrow$ FP16 instantly **doubles** FLOPs/byte; FP8 doubles throughput and halves memory again vs. FP16.
- Blackwell adds native **FP8 and FP4 Tensor Cores**, plus **hardware decompression**: weights stored compressed in HBM (even beyond FP4 — 4-bit/2-bit schemes) are expanded on the fly and cast to FP16/FP32 for accurate accumulation.
- Net: **effectively more memory bandwidth** — an architectural edge for memory-bound **token generation**.

The takeaway

Precision is a **bandwidth** optimization as much as a compute one: halve the bytes $\Rightarrow$ double the arithmetic intensity $\Rightarrow$ move toward the compute roof.

Profiling Workflow: Diagnose, Fix, Re-measure

- Memory-bound signature in **ncu**: **high DRAM utilization + low ALU utilization** — warps stalled on memory. Also watch **global load efficiency** dropping.
- **nsys** timeline shows idle stretches between kernels = GPU waiting on transfers.
- Expected overlap but see sequential copy-then-kernel? Two usual suspects: an unwanted **default-stream sync**, or a **missing pinned-memory buffer** — without pinned memory, `cudaMemcpyAsync` **cannot overlap** with kernels.
- Newer ncu niceties: **range replay**, source correlation, launch-stack size metric.
- After fixes: idle gaps vanish, **memory pipe utilization** climbs toward peak.

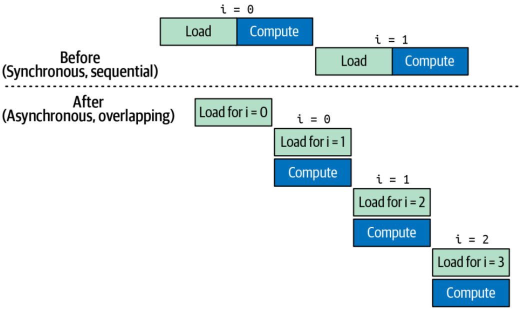


Figure 6-22. Sync (sequential) vs. async (overlapped) transfers + compute *Always measure after each change — profilers confirm whether stalls actually dropped.*

Key Takeaways (the Book's Eight)

- **SIMT**: 32-thread warps in lockstep; many warps in flight = latency hidden.
- **Hierarchy**: threads → blocks (≤1,024) → grids; minimize `__syncthreads()` barriers.
- **Occupancy vs. limits**: multiples of 32; per-SM: 64 warps, 32 blocks, 228 KB shared, 255 regs/thread.
- **Launch params**: 256 threads/block to start; grid = ceil(N/256); tune via profiling.
- **Async memory**: `cudaMallocAsync` + streams + pools; PyTorch's caching allocator ditto.
- **Memory ladder**: reuse in registers/shared/L1; coalesce into L2/HBM.
- **Unified Memory**: prefetch + advise to kill surprise stalls.
- **Roofline**: FLOPs/byte picks your battle; FP16/FP8/FP4 + hardware decompression move you right; **TMEM + UMMA** can shift kernels memory- → compute-bound.

Conclusion and What's Next

This chapter in one sentence

Make the GPU **busy** (occupancy), make data **close** (memory hierarchy), and let the **roofline** tell you which battle to fight next.

- The book's closing caveat: **max occupancy is not always optimal** — with enough **ILP** per thread, moderate/low occupancy can win; sometimes **fewer threads with more registers each** is faster. **Always benchmark.**
- Coming next in the book: warp divergence & memory-access tuning (Ch. 7–8), arithmetic intensity (Ch. 9), TMA & warp specialization (Ch. 10).
- My second talk — **Chapter 12** — builds on this: once single kernels are fast, how do we **orchestrate** them so the GPU never waits for the CPU?

Questions?

References & Further Reading

1. C. Fregly. *AI Systems Performance Engineering* (O'Reilly, 2026), Chapter 6.
2. NVIDIA. *Blackwell Tuning Guide* and *Blackwell Architecture Whitepaper* — SM limits, dual-issue schedulers, TMEM/tcgen05.
3. NVIDIA. *CUDA C++ Programming Guide* — thread hierarchy, Unified Memory, occupancy API, `__launch_bounds__`, stream-ordered allocation, PTX/fatbin.
4. S. Williams, A. Waterman, D. Patterson. *Roofline: An Insightful Visual Performance Model for Multicore Architectures*. CACM 52(4), 2009.
5. NVIDIA. *Compute Sanitizer User Guide* — memcheck, racecheck, initcheck, synccheck.
6. NVIDIA. *Nsight Systems* and *Nsight Compute* documentation — `nsys profile`, `ncu --section SpeedOfLight`, NVTX.
7. Book code and benchmarks: the book's GitHub repository (per-architecture results for the illustrative tables in this deck).